

- The problems below are taken from the Final collection. Each problem has exactly one correct answer among the four options (a)–(d). Clearly **select** your answer. Marks are +1 for a correct answer, $-1/3$ for a wrong answer and 0 for no answer.
- Properly **argue** your answers, with appropriate use of mathematics. Marks are +3 for a complete and correct explanation, +2 for a mostly complete and correct explanation, +1 for an explanation with significant gaps or errors but containing some correct reasoning, and 0 for an essentially incorrect explanation or one with no relevant correct content.
- Pay attention to the writing in your answers, since the **language use** is marked. The allowed language is English. The use of language is marked as +3 if all problems are answered in paragraphs with proper sentences, as +2 if there are some answers with limited use of language, as +1 if there are many answers with limited use of language, and as 0 if most answers have barely any sentences.
- The exam **grade** is computed according to the formula

$$\left(\frac{1}{3}Q + \frac{2}{3}D\right) \cdot \left(\frac{7+L}{10}\right),$$

where Q and D are respectively the averages of the answer marks and of the explanation marks, both normalized to 10, and L is the overall language mark.

- No electronic devices and no printed or written material are allowed.
- The exam duration is **2 hours**.

Name:

NIA or U:

1. A waveform $\phi(t)$ with similarity function $R_{\phi\phi}(\tau) = \text{sinc}(2\tau)$ is used to transmit information as

$$\sum_{i=-\infty}^{\infty} X_i \phi(t - iT),$$

where the X_i are symbols from an M -QAM constellation and T is the symbol period. The maximum rate without inter-symbol interference is

- (a) $2 \log_2 M$ bits per second.
 (b) $\frac{\log_2 M}{4}$ bits per second.
 (c) $\frac{\log_2 M}{2}$ bits per second.
 (d) $\log_2 M$ bits per second.
2. For some $a, b > 0$ with $a \neq b$, a modulation system maps two equiprobable independent bits (D_1, D_2) to the symbol

$$X = \begin{cases} a + jb & (D_1, D_2) = (0, 0) \\ a - jb & (D_1, D_2) = (0, 1) \\ jb & (D_1, D_2) = (1, 0) \\ a & (D_1, D_2) = (1, 1). \end{cases}$$

What is the average energy of the symbol X ?

- (a) $\frac{3a^2 + 3b^2}{4}$
 (b) $a^2 + b^2$
 (c) $2(a^2 + b^2)$
 (d) ab
3. What is the most likely output of the following MATLAB code?
- ```
n = 100000;
aux = rand(1,n);
xx(aux > 0.25) = 1j;
xx(aux < 0.25) = 3;
1/n*sum(xx.*conj(xx))
```
- (a) 3  
 (b) 5  
 (c) 2  
 (d) 7
4. A channel with input  $X \in \{0, 1\}$  and output  $Y \in \{0, 1\}$  has transition probabilities  $W(1|0) = \frac{1}{2}$  and  $W(0|1) = 0$ . What is the mutual information  $I(X; Y)$  when the input bits have probabilities  $\Pr(X = 0) = 1 - q$  and  $\Pr(X = 1) = q$ ? Here  $h_2(p)$  denotes the binary entropy function.
- (a)  $1 - q + h_2(\frac{1-q}{2})$   
 (b)  $q - 1 + h_2(\frac{1+q}{2})$   
 (c)  $q + h_2(\frac{1-q}{2})$   
 (d)  $-q + h_2(\frac{1+q}{2})$
5. We want to transmit reliably over a complex AWGN channel with bandwidth  $W = 2$  and noise spectral density  $N_0 = \frac{1}{4}$  at a rate of  $R = 4$  bits per second. This is possible if and only if the power  $P$  satisfies

6. A coded-modulation system sends one information bit  $D$  as the pair of symbols  $\mathbf{x}(D) = (X_1, X_2)$  with  $X_1 = X_2 = \sqrt{E_s} D$ , over the complex AWGN channel with noise variance  $\sigma^2 = 1$ . What is the set of all received pairs  $\mathbf{y} = (y_1, y_2)$  for which the two likelihoods coincide,  $W^2(\mathbf{y}|\mathbf{x}(0)) = W^2(\mathbf{y}|\mathbf{x}(1))$ ?

- (a)  $\text{Re}(y_1) = \text{Re}(y_2) = 0$   
 (b)  $\text{Re}(y_1) = \text{Re}(y_2) = \frac{\sqrt{E_s}}{2}$   
 (c)  $\text{Re}(y_1) + \text{Re}(y_2) = 0$   
 (d)  $\text{Re}(y_1) + \text{Re}(y_2) = \sqrt{E_s}$

7. A stream of bits interleaves control and information data so that 1 bit out of every 12 carries control commands. The stream is encoded with a channel code of rate  $\frac{6}{7}$ , and then groups of 7 coded bits are mapped onto sequences of  $n$  symbols of a 5-PAM constellation, with  $n$  the smallest length for which the mapping is reversible. What is the information rate of the system, in information bits (excluding control bits, before encoding) per modulation symbol?

- (a)  $\frac{6}{55}$   
 (b)  $\frac{3}{2}$   
 (c)  $\frac{11}{8}$   
 (d) 2

- (a)  $P < \frac{3}{5}$   
 (b)  $P > \frac{3}{5}$   
 (c)  $P > \frac{1}{5}$   
 (d)  $P < \frac{1}{2}$